

Association between maternal perceived stress during pregnancy and offspring DNA methylation changes in HPA axis genes at birth in the ECHO Consortium

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Abstract

Evidence has linked maternal exposure to stress during pregnancy with poor offspring health and neurodevelopmental outcomes. However, the precise mechanism by which this may occur has not been fully elucidated. In this study, we examined whether maternal perceived stress during pregnancy was associated with newborn blood DNA methylation (DNAm) in hypothalamic–pituitary–adrenal axis-related genes (NR3C1, FKBP5, and HSD11B2) in single CpG site and gene-based analyses. We analysed a subset of 661 mother–child pairs from the Environmental Influences on Child Health Outcomes cohort study that met our analytic inclusion criteria. Maternal perceived stress was measured during pregnancy using the perceived stress scale, and newborn DNAm was measured using the Illumina 450K and EPIC Beadchips in cord blood and dried blood spots. Single-site associations were evaluated using linear regression models, and gene-based associations were evaluated using mean burden and variance component tests, adjusted for sociodemographic and lifestyle covariates. Sex-stratified models were used to evaluate sex differential effects. Prenatal perceived stress was statistically significantly associated with newborn DNAm in one CpG site (cg06613263) in NR3C1 and with aggregate DNAm in NR3C1 and FKBP5. Aggregate DNAm in FKBP5 was more strongly associated with prenatal perceived stress in female infants. These results may have important implications for improving offspring health and well-being by providing molecular targets that can be used to identify high-risk individuals and as a basis for developing and evaluating effective behaviour and pharmaceutical interventions.

Keywords: perceived stress; pregnancy; DNA methylation; HPA axis

Introduction

Exposure to prenatal stress is a concerning public health problem among women in the USA. Data from the Pregnancy Risk Assess-

ment Monitoring System shows that 70% of women reported experiencing at least one stressful life event (SLE) in the year prior to their infant's birth, and prior studies have reported high rates of

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psychosocial stress among pregnant women in the USA [1–3]. Maternal stress exposure during pregnancy has been linked to various offspring health outcomes, including preterm birth, low birth-weight, sleep disorders, and neurodevelopmental disorders [4–8]. However, the underlying biological mechanisms by which these associations may occur are not fully characterized [9, 10]. Understanding these mechanisms may provide a means of identifying at-risk individuals or to facilitate the development and evaluation of interventions that mitigate the potential harms of maternal stress exposure during pregnancy.

Foetal DNA methylation (DNAm) may be an important biological mechanism in the link between prenatal stress and offspring health and development. The epigenome acts as an interface between genes and the environment, and DNAm is among the most extensively studied forms of epigenetic modification [7, 11–14]. DNAm involves the reversible addition or removal of a methyl group to the cytosine nucleotides in the CpG sites of a gene, leading to increases or decreases in gene expression. The gestational period is one of large-scale, dynamic DNAm change, and this developmental window is marked by a heightened sensitivity to environmental stimuli, leading to changes in DNAm state [11]. In a process known as foetal programming, these DNAm changes lead to functional and organizational changes in various biological systems, including the central nervous, autonomic nervous, neuro-endocrine, and immune systems [7, 11]. Foetal programming may prime the individual for future environmental insults and also may lead to an increased susceptibility to adverse health outcomes across a range of life stages such as preterm birth, neuropsychiatric disorders in childhood, and chronic diseases in adulthood [7, 11, 15]. Indeed, DNAm changes have been associated with preterm birth and neurodevelopmental conditions [16–18]. It is important to consider these factors in analyses, as we do here, that examine prenatal perceived stress exposure associations with DNAm patterns in offspring.

The foetal hypothalamic–pituitary–adrenal (HPA) axis is likely an important target of these processes. The HPA axis plays a critical role in stress response and glucocorticoid regulation, and the foetal HPA axis has shown marked sensitivity to the intrauterine environment [7, 10, 11]. Maternal prenatal experiences of stress marked by persistently elevated maternal glucocorticoid levels create dysregulation of the foetal HPA axis, including greater offspring cortisol response to an environmental stressor and changes in basal cortisol levels, through a variety of processes, including DNAm changes in HPA axis-associated genes in the foetus [9, 10, 19, 20]. HPA dysregulation has in turn been linked to neuropsychiatric outcomes, including attention deficit hyperactivity disorder and depression in children and adolescents [21, 22].

A growing number of studies have examined associations between maternal stress during pregnancy and newborn DNAm in HPA axis-associated genes, using both single CpG site and gene-based methylation analyses [7, 23]. These studies have almost exclusively examined, and have largely found significant associations between, the Nuclear Receptor Subfamily 3 Group C Member 1 (NR3C1) gene, which is critical to HPA functioning by controlling the amount of cortisol in the body, and maternal stress during pregnancy [7, 23–28]. Comparatively, only a few studies have examined other relevant genes implicated in foetal HPA axis functioning and overall stress reactivity, including hydroxysteroid 11-beta dehydrogenase 2 (HSD11B2), which converts cortisol to inactive cortisone [29], and FK506-binding protein 5 (FKBP5), which modulates glucocorticoid receptor activity in response to stress [30]. Additionally, these studies have reported largely inconsistent results likely due to several factors, including small study sizes,

choice of covariates, and choice of prenatal stress measure [31–34].

Much of this prior research has been conducted in small study populations with little racial, ethnic, or socioeconomic diversity. Many of these studies examined fewer than 200 parent–child pairs, and the largest to date examined 482 dyads. This introduces issues related to power and the generalizability of study results [27, 31]. Much of this prior work has also focused on self-report measures of objective stressors, and related psychopathology, such as maternal depression and anxiety, rather than measures of perceived stress or distress [24–26, 33, 35]. Perceived stress, however, may be a more useful measure of maternal stress than SLEs in these investigations. In proposed models of the stress process, perceived stress is considered more proximal to the body's peripheral stress response and so may be more predictive of illness outcomes than SLEs [36, 37]. This is because the pathogenic effects of stressful events likely stem from the emotional, subjective response to the event, rather than from the event itself [38]. Furthermore, the same event can lead to differences in perceived stress between individuals with different personal or social contexts, including social support, family structure, and cultural values and norms, and as such measures of objective stress may imperfectly capture the stress response and its impact on foetal development [38].

Additionally, as stated previously, most such studies have prioritized DNAm in NR3C1 genes, limiting our understanding of the effect of prenatal maternal stress on DNAm across various HPA axis-relevant genes. Sex-specific effects are also largely understudied in these investigations, despite evidence of sex differences in foetal development and offspring susceptibility to adverse outcomes due to prenatal maternal stress exposure [7, 23]. Finally, studies that have examined the association between maternal stress during pregnancy and offspring gene-level DNAm have primarily used burden tests that use mean or median methylation levels to capture aggregate DNAm across the region of interest [31]. These methods facilitate gains in power by removing the need to correct for multiple testing as in single-site analyses. Additionally, as individual CpG sites are unlikely to act alone in regulating gene expression, aggregate methods may identify more functionally relevant changes in DNAm than single CpG analysis. However, these methods may fail to account for different directions and magnitude of effect among CpG sites, thus resulting in loss of power and a potential masking of associations [39].

In the current study, we aim to address these limitations using a large, racially and ethnically diverse, US population-based cohort by examining the association between maternal stress during pregnancy and neonatal blood DNAm in the HPA axis-associated genes, NR3C1, HSD11B2, and FKBP5, as well as sex-specific effects through both site and gene-based analyses using flexible and computationally efficient variance component tests like the sequence kernel association test (SKAT) primarily used in rare genetic variant analyses to facilitate gains in power [39–41]. Unlike the aggregate methods described previously, SKAT models methylation at CpG sites more flexibly across a gene, allowing different directions and magnitudes of effect at each site in a gene. This flexibility is advantageous when examining gene-level DNAm changes associated with an exposure of interest because methylation levels influence downstream gene expression differently depending on its location in a gene. For example, methylation at CpGs located in a gene promoter region typically decreases gene expression, and methylation at CpGs in the gene body typically increases gene expression. Therefore, by facilitating different direction and magnitudes of effect between sites, SKAT overcomes as-

assumptions of simpler aggregate methylation tests that each CpG methylation site in a gene contributes equally to an outcome and in the same direction, which does not align with biologic expectations. In these analyses, we use data from the Environmental Influences on Child Health Outcomes (ECHO) Program and focus on prenatal perceived stress as a measure of psychological distress to capture the subjective, cognitive appraisal of life stress. We hypothesize that higher levels of maternal perceived stress during pregnancy will be associated with increases in newborn blood DNAm levels in *NR3C1*, *HSD11B2*, and *FKBP5* genes and that effects will differ by child's sex.

Methods

Overall study design

The ECHO Program is a multisite cohort study comprised of pregnancy and pediatric sites designed to understand how environmental factors from preconception to early childhood influence child health and development, with the goal of minimizing disease risk and promoting positive health and well-being [42]. The ECHO Program includes a diverse cohort of participants from all US states, the District of Columbia, Puerto Rico, and Native American Tribal Nations (see Knapp et al. [42] for more details) [43, 44].

Participants

To examine the association between prenatal perceived stress and newborn blood DNAm, we included a subset of ECHO participants in our analyses that met the following inclusion criteria: (1) the mother reported on perceived stress during pregnancy, (2) the pregnancy was a singleton gestation, and (3) the child had a high-quality (passed quality control; QC) blood-derived (umbilical cord blood or blood spot) DNAm measure on either the Illumina Infinium HumanMethylation450 or Infinium MethylationEPIC BeadChips. The final analytic sample consisted of 661 mother-child dyads (Supplementary Fig. 1) from four ECHO sites: Healthy Start, Autism Spectrum Disorder Enriched Risk Cohort (ASDER), Atlanta ECHO Cohort of Emory, and Madres. Site-specific recruitment methods and population characteristics are outlined in Supplementary Table 1.

Perceived stress during pregnancy

Prenatal Perceived stress was measured using the perceived stress scale (PSS) [45]. The PSS is the most widely used measure of perceived stress in epidemiologic research and captures the degree to which events in one's life are considered stressful [38]. Two versions of the PSS were administered at ECHO sites included in these analyses: 10-item (PSS-10) and 14-item (PSS-14). Across scales, each item is scored on a 5-point Likert scale that ranges from never (0) to very often (4). Item response theory (IRT) was used to harmonize all available PSS data across participants to the perceived stress NIH toolbox T-score metric, which has a mean of 50 and a standard deviation of 10 in the general US population [46].

Some participants had multiple perceived stress T-score measurements during pregnancy, and prenatal perceived stress was measured between 8 and 38 weeks. As repeated measures were correlated in our sample (Pearson's $\rho = 0.63$, P -value $< .001$), we generated a mean T score from all available time points to examine the effect of maternal perceived stress across the entire prenatal period. We treated perceived stress as both a continuous and a binary variable to evaluate the robustness of our findings. For binary perceived stress scores, population norms derived from a representative US sample were used to delineate T-score cate-

gories (low to medium (binary category 0): <60 and high (binary category 1): ≥ 60) [47].

DNAm measures

Sites used approved site-specific protocols to collect umbilical cord blood or newborn blood spots. DNAm was measured using the Illumina Infinium HumanMethylation450 or Infinium MethylationEPIC BeadChips. Raw participant data obtained from each site were run through the ECHO Illumina QC pipeline using the 'minfi' R/Bioconductor package, and β values (ranging from 0 for completely unmethylated to 1 for completely methylated) calculated for each probe and sample. Samples were filtered for locus-level technical failures, sex discrepancy, poor bisulfite conversion, high detection P -values ($>1\%$ probes with $\text{detP} > .05$), and low-quality probes ($>1\%$ probes $\text{beadC} < 3$). Similarly, probes for which $>1\%$ samples with $\text{detP} > .05$ and $>1\%$ of samples with $\text{beadC} < 3$, as well as those that were cross reactive or had a single nucleotide polymorphism (SNP) at the CpG site were removed. This processing was followed by normalization procedures using the normal-exponential out-of-band correction method. Cell type proportions were estimated using biospecimen type appropriate reference panels [48]. A total of 661 samples (524 on the 450K array and 137 on the EPIC array) that met study inclusion criteria passed QC across four study sites (Supplementary Fig. 1). From the 430 739 probes on the 450K BeadChip and 733 758 on the EPIC BeadChip that passed QC, we derived a set of 71 CpG sites found on both arrays located in the *NR3C1*, *FKBP5*, and *HSD11B2* genes based on Illumina's annotation files for the 450K and EPIC BeadChips that assign each CpG site to a UCSC known gene. Separately, we derived a second dataset of 66 CpG sites that passed QC annotated to the *NR3C1*, *FKBP5*, and *HSD11B2* genes located on the EPIC array only to use in complementary analyses.

Covariates and effect modifiers

Covariates included maternal age at time of delivery, maternal education as an indicator for socioeconomic status, maternal prenatal depressive symptoms, prenatal alcohol consumption, maternal prenatal tobacco use, child's sex, and cell type proportions. Maternal prenatal depressive symptoms, prenatal alcohol consumption, and maternal prenatal tobacco use may be mediators of the association between prenatal perceived stress and newborn DNAm. We add these factors as covariates in our models to examine the direct effect of perceived stress independent of these factors. We also analysed child's sex as an effect measure modifier to evaluate sex-specific effects and examined the effects of preterm birth on results in sensitivity analyses. Notably, we do not adjust for ancestry principal components in our analyses as rather than conduct an epigenome-wide screen, we focus our analyses on three genes. As a result, there is less of a concern of confounding due to population stratification. Additionally, we do not adjust for self-reported race/ethnicity, as we do not expect these socially defined factors to confound the association between perceived stress during pregnancy and newborn blood DNAm.

Covariate information was obtained from demographic forms, registration data, medical record abstraction, lifestyle forms, and validated scales. Prenatal maternal depressive symptom scores were measured using the PROMIS v1.0—Depression 8a scored on the PROMIS-score metric (mean = 50, standard deviation = 10) [49]. Scores from other depression scales, including the Edinburgh postnatal depression scale, were converted to the PROMIS-score metric using crosswalk tables to facilitate the merging of data across studies as described previously [50]. Maternal alcohol con-

sumption and tobacco use during pregnancy were assessed by self-report through ECHO lifestyle forms. Cell type proportions were estimated as described above. Finally, preterm birth was delineated as birth at ≥ 37 weeks' gestation (binary category = 0) and birth at < 37 weeks' gestation (binary category = 1).

Statistical analysis

We conducted our analyses in R version 4.2.2 [51]. We calculated descriptive statistics of mother and child characteristics for the analytic sample using means ($\pm$ SD) and proportions (%) where appropriate, overall and stratified by low-medium vs. high prenatal perceived stress. We imputed missing covariate data using random forest ('missForest' package in R) [52].

DNAm β -values (which represent the proportion of methylated sites in bulk tissues; $0 < \beta < 1$) were $\log_2$ transformed to M-values for use in primary statistical analyses. We also conducted complementary analyses using DNAm β -values to estimate % change in DNAm to aid in interpretation.

Analyses were conducted in participants with 450K and EPIC BeadChip measures, separately and then meta-analysed. We used the *comBat()* function from the 'sva' R/Bioconductor package [53] to correct for known batch effects. Using the *lmFit()* function from the 'limma' R/Bioconductor package [54], we then used multivariate linear regression models to analyse the association between prenatal perceived stress treated as both a continuous and binary variable, and DNAm measures at each CpG site under study (71 overlapping sites and 66 EPIC array only sites), adjusting for maternal age at time of child's birth, maternal depressive symptoms, maternal prenatal alcohol consumption, maternal prenatal tobacco use, maternal education, and child's sex. We combined group-specific results through a fixed effects inverse variance weighted meta-analysis using the METAL software [55]. To investigate sex-specific effects, we also conducted our analyses in female-only and male-only samples and assessed statistical interaction between prenatal perceived stress and child sex. To account for multiple testing, we used a Bonferroni-corrected significance threshold of 0.05/number of sites.

Sensitivity analysis

We carried out sensitivity analyses to evaluate the impact of several factors on the results. To examine the effect of preterm birth, we re-ran analyses restricting to children born ≥ 37 weeks' gestation as infants born preterm may have different DNAm profiles than infants carried to term [16, 17]. Additionally, we re-ran analyses to exclude ASDER, an autism spectrum disorder familial enriched site, to evaluate whether results were impacted by familial ASD risk. Finally, given the expected correlation between perceived stress during pregnancy and maternal depressive symptoms, we used Pearson correlation to evaluate the bivariate correlation between PSS and maternal prenatal depression and conducted sensitivity analyses in which we excluded maternal depressive symptoms from our models.

Gene-based tests

We conducted gene-based tests of the association between continuous prenatal perceived stress and neonatal blood DNAm in four regions using CpG sites shared between arrays and EPIC only CpG sites: (1) NR3C1 gene region, (2) FKBP5 gene region, (3) HSD11B2 gene region, and (4) all three genes together. Genes were delineated using Illumina's annotation file for the 450K and EPIC BeadChips. Analyses were conducted in the full, female-only and male-only samples.

Mean DNAm level tests. Mean methylation levels were calculated across CpG sites in each of the four gene regions. We employed generalized linear regression models to regress mean methylation levels in each region onto prenatal perceived stress, adjusting for covariates.

SKAT. We used the SKAT function in the 'SKAT' R package to test for the cumulative effect of methylation across multiple sites simultaneously using the SKAT method [39]. SKAT is a computationally efficient variance component score test that evaluates the effect of multiple variants in a region on a phenotype [39]. Though traditionally used in rare variant associations studies, prior work has adopted the method for use in gene-based methylation tests due to improved power and accommodation of different directions of effect [40, 41]. We used four sets of SKAT parameters:

1. (Primary) Linear kernel with equal weights (each site assigned a weight of 1).
2. Quadratic kernel with equal weights (each site assigned a weight of 1).
3. Linear kernel with sites weighted based on the effect estimate obtained in individual site analyses ($1 + \text{abs}(\text{effect estimate})$).
4. Quadratic kernel with sites weighted based on effect size obtained in individual site analyses ($1 + \text{abs}(\text{effect estimate})$).

In secondary analyses, we reran SKAT using the SKAT-O or optimal test method to identify the optimal linear combination between SKAT and mean burden tests [56].

As above, we conducted analyses in participants with 450K and EPIC array measures separately, and the resulting effect estimates and P-values combined by meta-analysis. For gene-based tests, we used a significance threshold of $P < .05$.

Results

Analytic sample descriptive characteristics

A total of 661 mother-child dyads from four ECHO cohort sites met our inclusion criteria (Supplementary Table 1). In our analytic sample, ~10% of mothers reported high perceived stress levels (T-score ≥ 60) (Table 1). This proportion ranged from 3.6% at the MADRES cohort site to 12.5% at the ASDER cohort site (Supplementary Table 2). These mothers were less likely to identify as White (68% vs. 75%), less likely to identify as Hispanic (28% vs. 36%), more likely to report alcohol consumption (20% vs. 17%) and tobacco use (15.4% vs. 6.5%) during pregnancy, and they also had higher depressive symptom scores (mean = 54.48, SD = 5.52 vs. 46.20, SD = 7.58) than women who reported low or medium levels of prenatal perceived stress.

CpG site analysis

We used adjusted linear regression models to evaluate the association between continuous and binary prenatal perceived stress and DNAm measured in newborn blood samples at each of the 71 CpG sites located in the target genes measured on both the EPIC and 450K platforms (Supplementary Table 3; volcano plot is shown in Fig. 1). We observed suggestive associations ($P < .05$) at seven CpG sites located in the NR3C1 and FKBP5 genes (Table 2). At the top site, prenatal perceived stress levels were significantly ($P < .0007$) associated with blood DNAm at probe cg06613263 annotated to the NR3C1 gene, located in the gene body in the North Shore region within 2 kb upstream of the CpG island (Table 2). We observed a 0.021% increase in DNAm for every one-unit increase in prenatal perceived stress (P -value = .00027).

Table 1. Descriptive characteristics of parent and child samples.

Covariate	Prenatal perceived stress ^a			P-value
	Total (N = 661)	Low to medium perceived stress (N = 596)	High perceived stress (N = 65)	
Maternal self-reported race %				
Black	110 (16.6)	97 (16.3)	13 (20.0)	.098
White	492 (74.4)	448 (75.2)	44 (67.7)	
Other	59 (8.9)	51 (8.6)	9 (13.8)	
Missing	2 (0.3)			
Self-reported Hispanic ethnicity (%)				
Hispanic	233 (35.2)	215 (36.1)	18 (27.7)	.228
Missing	1 (0.2)			
Maternal age at delivery				
Mean (SD)	28.10 (6.01)	28.26 (6.07)	26.57	.034
Missing	0		(5.32)	
Gestational weeks at delivery				
Mean (SD)	38.97 (1.36)	38.98 (1.38)	38.88	.152
Missing	0		(1.17)	
Preterm birth (%)	29 (4.4)	28 (4.7)	<5 (<5)	.389
Missing	0			
Child sex (%)				
Female	325 (49.2)	295 (49.5)	30 (46.2)	.703
Missing	0			
Maternal depressive symptoms during pregnancy				
Mean (SD)	47.02 (7.80)	46.20 (7.58)	54.48	<.001
Missing	15 (2.3)		(5.52)	
Prenatal alcohol consumption (%)				
Yes	113 (17.1)	100 (16.8)	13 (20.0)	.630
Missing	41 (6.2)			
Prenatal tobacco use (%)				
Yes	49 (7.4)	36 (6.5)	10 (15.4)	.020
Missing	0			
Maternal education (%)				
Less than high school	117 (17.7)	104 (17.4)	13 (20.0)	.877
High school degree, GED or equivalent	137 (20.7)	124 (20.8)	13 (20.0)	
Some college, no degree, and above	407 (61.6)	368 (61.7)	39 (60.0)	
Missing	3 (0.5)			

^aHigh perceived stress was defined as having a PSS T score of equal to or greater than 60. Medium/low was defined as having a PSS value below 60.

Cg06613263 also emerged as the top site when perceived stress was treated as a binary variable (Supplementary Table 4). We observed a 0.27% (P-value = .011) increase in DNAm at cg06613263 in newborns whose mothers had high perceived stress levels over those whose mothers had low or medium perceived stress levels (Table 2).

We also performed analyses to evaluate the association between maternal perceived stress during pregnancy and DNAm in each of the 66 CpG sites specific to the EPIC array (Supplementary Table 5). Eight of these were nominally associated with continuous prenatal perceived stress T score levels (Table 2). We observed decreased neonatal blood DNAm with increased prenatal perceived stress at all but the top site where blood DNAm increased with increased perceived stress (cg24052866 annotated to NR3C1, % change = 0.059%, P-value = .0015). However, none of these associations met a conservative Bonferroni threshold correcting for multiple testing.

In sex-stratified analyses, cg06613262 was also the top site in male only models where we observed a nominally significant association between DNAm and prenatal perceived stress (% difference = 0.028% per one-unit increase in prenatal per-

ceived stress, P-value = .0029) (Supplementary Table 6). In female only models, we similarly observed a suggestive association at this site (% difference = 0.014% per one-unit increase in prenatal perceived stress, P-value = .031). In statistical interaction models, we found no evidence of interaction by sex at this site ($P_{\text{interaction}}$ value = .33). However, the lowest P-value was observed at cg07724674 annotated to the HSD11B2 gene (% difference = 0.021% per one-unit increase in prenatal perceived stress, P-value = .00096) (Supplementary Table 7). In male children, the association between maternal perceived stress during pregnancy and newborn DNAm was not significant and showed an opposite direction of effect at this site (% difference = -0.0069% per one-unit increase in prenatal perceived stress, P-value = .358). Additionally, statistical interaction models showed evidence of interaction by sex at this site ($P_{\text{interaction}}$ value = .069).

Sensitivity analysis

Sensitivity analyses were carried out to examine the effect of preterm birth or increased familial predisposition to autism on the association between prenatal perceived stress and newborn blood DNAm at cg06613262. These results are presented in

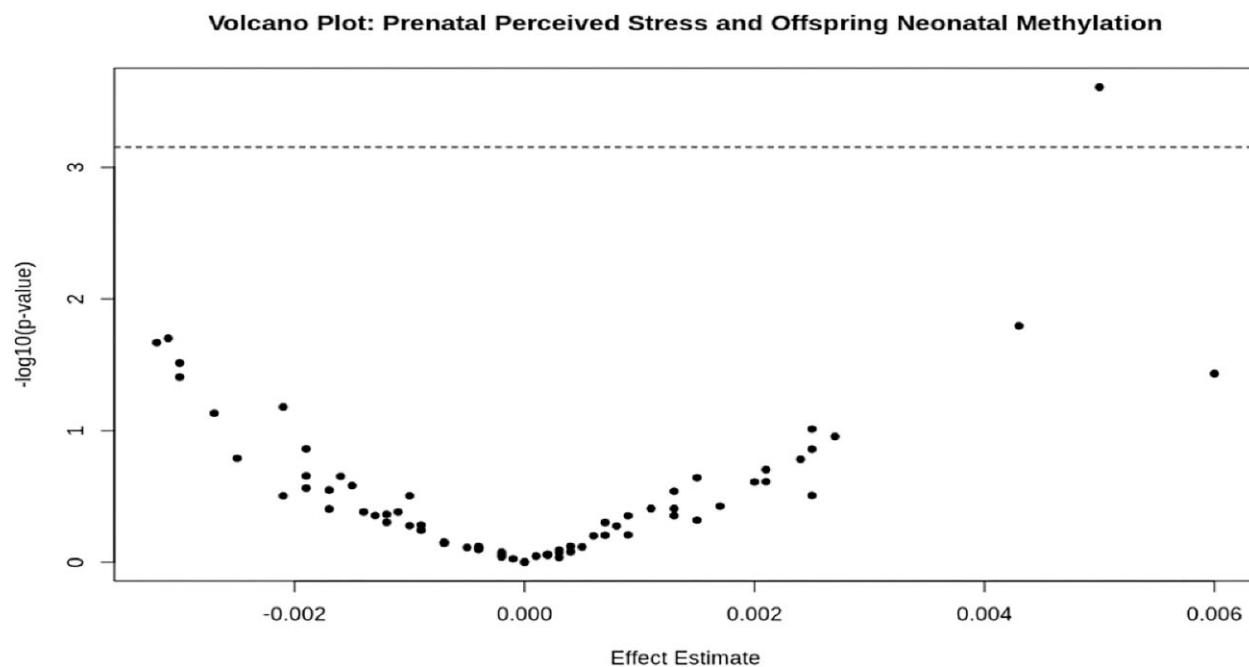


Figure 1. Volcano plot of association testing results for prenatal perceived stress and newborn blood DNAm in NR3C1, FKBP5, and HSD11B2. Each point represents a CpG site. The dashed line represents the Bonferroni-corrected significance threshold ($P < .0007$).

Table 3. Across all models, effect estimates did not vary largely from that obtained in the main, fully adjusted model, though the model that excluded participants from an ECHO site that enrolled participants with high familial autism predisposition produced a smaller % difference in blood DNAm with increased perceived stress (% DNAm change per one-unit increase on perceived stress = 0.014%, P -value = .012) compared to the full model.

Additionally, we conducted sensitivity analyses to evaluate the effect of removing maternal prenatal depressive symptoms from our models assessing the association between prenatal perceived stress and newborn blood DNAm at cg06613262 (Table 3). We found a positive correlation between prenatal perceived stress and maternal prenatal depressive symptoms ($r = 0.38$, P -value < .001). Excluding maternal prenatal depressive symptoms from our models resulted in a smaller % difference in newborn blood DNAm with increased perceived stress (% DNAm change per one-unit increase on perceived stress = 0.007% P -value = .009) compared to the fully adjusted model, though the direction of effect remains consistent.

Gene-level tests

Mean methylation burden tests and SKAT were used to examine the association between prenatal perceived stress and offspring DNAm in each of four regions: (1) NR3C1, (2) FKBP5, (3) HSD11B2, and (4) all three genes together.

Mean methylation burden test

We calculated mean methylation for each participant within each gene region outlined above using the set of overlapping CpG sites between the 450K and EPIC arrays. A breakdown of each region and corresponding CpG sites is shown in Supplementary Table 8. We regressed mean DNAm in each region onto continuous perceived stress using fully adjusted linear regression models and found no evidence of association (Table 4). Models using mean

methylation values calculated from EPIC-only CpG sites similarly returned null results (Supplementary Table 9).

SKAT

We then used SKAT, applying various kernel and weight combinations, to more flexibly model the association between aggregate newborn blood DNAm across overlapping 450K and EPIC CpG sites within each delineated region and prenatal perceived stress. These results are presented in Table 5. First, we used a linear kernel to model the additive effects of each CpG site and weighted all sites evenly (weights = 1 or unweighted). We observed a significant association between prenatal perceived stress and DNAm patterns in the NR3C1 (P -value = .040) and FKBP5 genes (P -value = .024), and across all 71 sites in the three HPA axis pathway genes (NR3C1, FKBP5, and HSD11B2) together (P -value = .022). Secondary analyses in which we applied a quadratic kernel to model potential interactions between sites yielded similar results, showing significant effects in NR3C1 (P -value = .040) and FKBP5 (P -value = .037), and the set of all sites across the three genes under study (P -value = .039). Weighting the models by effect estimates obtained in single-site analyses had no effect on the associations. Similar results were obtained when we restricted these analyses to the set of 66 CpG sites measured on the EPIC array only (Supplementary Table 10).

Finally, to optimize between SKAT and burden test approaches, we reran SKAT using the SKAT-O or optimal test method using a linear kernel and weighting all sites equally (Supplementary Table 11). Resulting P -values were larger in all four regions compared with that produced in original SKAT analyses, with associations between prenatal perceived stress and DNAm in FKBP5 and across all sites only approaching significance (P -value_{FKBP5} = .055, P -value_{All sites} = .076). Correlations between CpG sites within each region were 0 for all but the HSD11B2 region where the correlation between sites was determined to be 0.75.

Table 2. Association between prenatal perceived stress and newborn blood DNAm levels at $P < .05$.

	Probe ID	Gene symbol	Effect ^a	SE ^a	P-Value ^a	BH-FDR adjusted P-value ^a	% DNAm difference ^a
EPIC and 450K sites ($n = 71$ CpGs measured); $N = 661$ samples							
Continuous PSS		NR3C1	0.0050	0.0014	.00027	.019	0.021%
	cg06613263	FKBP5	0.0042	0.0018	.016	.408	0
	cg00610228	NR3C1	− 0.0033	0.0014	.020	.408	−0.0069
	cg18849621	NR3C1	− 0.0030	0.0013	.025	.408	0
	cg18019515	FKBP5	− 0.0030	0.0014	.030	.408	−0.0069
	cg00862770	FKBP5	0.0061	0.0029	.035	.408	0.069
	cg06087101	NR3C1	− 0.0030	0.0014	.040	.409	0
	cg14558428	NR3C1	0.074	0.029	.011	.484	0.27
	cg06613263	NR3C1	0.086	0.035	.014	.484	0.86
	cg07733851	HSD11B2	− 0.063	0.028	.024	.576	−0.042
Binary PSS (65 high PSS group)	cg16545496	NR3C1	− 0.061	0.031	.051	.781	−0.15
	cg18849621						
	EPIC only sites (66 CpGs measured); $N = 137$ samples						
	Continuous PSS						
		NR3C1	0.0082	0.0026	.0015	.102	
	cg24052866	NR3C1	− 0.0061	0.0025	.015	.316	0.05910886
	cg25579735	FKBP5	− 0.0068	0.0029	.021	.316	−0.0228306
	cg06409316	NR3C1	− 0.0058	0.0025	.022	.316	−0.0453399
	cg24801588	NR3C1	− 0.0040	0.0019	.032	.316	0.04393455
	cg13514002	FKBP5	0.0052	0.0024	.033	.316	−0.0509034
	cg07633853	FKBP5	− 0.036	0.017	.034	.316	−0.013942
	cg21517946	NR3C1	− 0.0043	0.0021	.043	.556	0.03 516 776
	cg14438279						−0.1492263

^aAdjusted for maternal age at delivery, maternal depression, prenatal alcohol consumption, prenatal tobacco use, maternal education, and child's sex. Note: The coefficients and P-values were generated based on methylation M-value. %DNAm difference was generated based on methylation beta value.

Table 3. Sensitivity analyses showing the association between prenatal perceived stress and newborn blood DNAm levels, at probe cg06613263 (NR3C1).

Analysis	Sample size	Effect	Standard error	P-value	BH-FDR adjusted P-value ^a	% DNAm difference
Main analysis	661	0.0050	0.0014	.00027	.019	0.021%
≥37 weeks gestation	632	0.0045	0.0015	.0025	.199	0.021%
Excluding autism cohort site	637	0.0038	0.0015	.012	.257	0.014%
Excluding maternal prenatal depression from statistical model	661	0.0027	0.0010	.009	.312	0.007%

^aAdjusted for maternal age at delivery, maternal depression, prenatal alcohol consumption, prenatal tobacco use, maternal education, and child's sex. Note: The coefficients and P-values were generated based on methylation M-value. %DNAm difference was generated based on methylation beta value.

In sex-stratified analyses, aggregate newborn blood DNAm in the FKBP5 gene remained statistically significantly associated with prenatal perceived stress in female-only samples (P -value = .004), but not in male-only samples ($P = .60$) (Table 6).

On the other hand, the association between DNAm in HSD11B2 and prenatal perceived stress showed stronger significance in male newborns (P -value = .064), compared to female newborns (P -value = .26).

Table 4. Prenatal perceived stress and newborn mean methylation from shared CpG sites association testing results.

Gene symbol	Number of sites	β (95% confidence interval) ^a	P-value ^a
NR3C1	33	−0.0001 (−0.0010, 0.0007)	.79
FKBP5	26	0.0001 (−0.0008, 0.0010)	.81
HSD11B2	12	0.0004 (−0.0008, 0.0015)	.52
Overall	71	0.0000 (−0.0007, 0.0007)	.97

^aAdjusted for maternal age at delivery, maternal depression, prenatal alcohol consumption, prenatal tobacco use, maternal education, and child's sex.

Table 5. Continuous prenatal perceived stress and newborn DNAm gene-based SKAT results.

Gene symbols	Number of sites	Linear unweighted		Quadratic unweighted		Linear weighted		Quadratic weighted	
		Q	P-value	Q	P-value	Q	P-value	Q	P-value
NR3C1	33	609.68	.040	904 900.60	.040	609.68	.040	904 900.60	.040
FKBP5	26	613.45	.024	365 607.1	.037	613.45	.024	365 607.1	.037
HSD11B2	12	151.94	.14	63 388.65	.17	151.94	.14	63 388.65	.17
All sites	71	1375.07	.022	368 891.00	.039	1375.07	.022	368 891.00	.039

Note: Adjusted for maternal age at delivery, maternal depression, prenatal alcohol consumption, prenatal tobacco use, maternal education, and child's sex.

Table 6. Continuous prenatal perceived stress and newborn DNAm SKAT results (linear, unweighted) in female-only and male-only samples.

Region	Number of sites	Female samples (N = 325)		Male samples (N = 336)	
		Q	P-value	Q	P-value
NR3C1	33	107.26	.53	286.12	.22
FKBP5	26	345.45	.0042	153.66	.60
HSD11B2	12	46.21	.26	107.68	.074
All sites	71	498.92	.080	547.45	.28

Note: Adjusted for maternal age at delivery, maternal depression, prenatal alcohol consumption, prenatal tobacco use, and maternal education.

Discussion

Our findings contribute to the growing body of literature investigating the molecular processes involved in foetal response to maternal stress exposure during pregnancy. Prenatal stress has been linked to poor offspring physical and neurodevelopmental outcomes, though the underlying mechanisms remain unclear [7, 8]. Prior work has identified the HPA axis as a pathway of interest, and studies have examined associations between prenatal stress, largely focused on SLEs, and stress-related psychopathology like depression and anxiety. To our knowledge, this is one of the largest studies to date in a diverse US population that has examined the association between prenatal perceived stress and newborn DNAm in HPA axis-related genes (NR3C1, FKBP5, and HSD11B2) and one of the few that have examined potential sex differential effects in offspring.

In single-site analyses, we found evidence that higher prenatal perceived stress exposure on a continuous scale was associated with DNAm changes in various sites located in NR3C1 and FKBP5 genes. Specifically, maternal perceived stress during pregnancy was significantly associated with increased DNAm in cg06613263 located in the NR3C1 gene body in a CpG North Shore region. One prior study has identified an association between maternal financial difficulties at 8 months postnatal and offspring DNAm at this site at age 7 [57]. Similarly, when prenatal perceived stress was modelled as a binary exposure, high perceived stress during pregnancy was associated with increased offspring DNAm, com-

pared to low and medium perceived stress, though this association was not significant after correcting for multiple testing. These results largely corroborate previous findings. Prior studies have reported associations between prenatal stressors (such as war-related stress, maternal depression, and anxiety) and newborn cord blood and placenta DNAm in NR3C1 CpG sites [24, 25, 32, 58]. A few studies, however, failed to find significant associations between prenatal maternal stress and newborn DNAm in this gene [26, 27]. Mansell et al. [27], e.g. found that while prenatal perceived stress was marginally associated with site-specific methylation in NR3C1 in infant cord blood, this association did not remain after correcting for multiple testing. This discrepancy in results may reflect lack of statistical power due to small sample sizes, differences in sites assessed and DNAm measurement methods, and differences in choice of covariates across studies.

In addition to single CpG analyses, a complementary gene-level test using both mean methylation burden tests and the SKAT method. SKAT facilitates a flexible modelling approach, overcoming the assumption that methylation at multiple CpGs shows similar magnitudes and directions of effect used in standard aggregate methylation tests. Single-site analyses provide high-resolution information of specific positions in the genome that are different among newborns of mothers who experienced different levels of perceived stress during pregnancy, which may be useful for downstream *in vitro* functional analyses and summary statistic gene-level meta-analytic approaches across future studies, e.g. comb-p [59]. Gene-level analyses provide a complementary ap-

proach because they can improve power to detect significant associations, given their reduced multiple testing burden compared to single CpG analyses, and because focusing on gene-level targets may provide broader mechanistic insight to inform gene-level intervention targets.

The two region-based methods employed in this study produced inconsistent results. In SKAT region-based analyses, we observed a significant association between prenatal perceived stress and newborn DNAm in *NR3C1*, *FKBP5*, and across the three genes together. However, mean methylation burden tests produced null results across all regions. Additionally, while single CpG site analyses identified statistically significant associations between prenatal perceived stress and newborn DNAm in *NR3C1*, there were no statistically significant sites in *FKBP5*. Monk et al. [31] showed that prenatal distress was associated with increased *NR3C1*, *FKBP5*, and *HSD11B2* placental mean methylation using bivariate Spearman correlation, though failed to adjust for relevant covariates, introducing concerns about unmeasured confounding.

The discrepancy in results obtained in the current study likely stems from test characteristics. Mean methylation burden tests assume the same direction and a similar magnitude of effect across CpG sites, and so may result in loss of power and masking of associations [39]. However, variance components score tests, such as SKAT, allow for more flexible modelling of the aggregate effect of CpG sites in a region by assuming that each site is independent, resulting in gains in power and improved detection when sites are truly independent [39, 40, 56]. Therefore, our results suggest that sites across each region act independently with different directions and magnitudes of effect. Sensitivity analysis using the SKAT-O method substantiates these conclusions. The SKAT-O or optimal test creates the optimal linear combination of SKAT and burden tests to optimize power by determining and incorporating correlation between sites into statistical models [56]. In our analyses, SKAT-O results showed no correlation ($\rho = 0$) in *NR3C1* and *HSD11B2* genes, indicating the SKAT method was optimal. This is in keeping with what we would expect biologically as CpG sites have different downstream effects that depend in part on where the site is located in the gene. For example, DNAm in the promoter region has been shown to be associated with transcriptional repression, and DNAm in gene bodies has been shown to result in transcriptional activation [60]. Furthermore, our finding that prenatal perceived stress was statistically significantly associated with newborn DNAm in *FKBP5* in SKAT gene-level but not in single CpG site analysis may be attributed to power gained through this SKAT method.

Additionally, SKAT analyses using linear and quadratic kernel functions both revealed significant associations in *FKBP5*, suggesting both additive and synergistic effects between sites. However, given the smaller *P*-value obtained in the linear kernel model, there is greater statistical evidence in support of additive effects within these regions.

In sex-stratified analyses, the top site in female-only samples was located in *HSD11B2* single-site analyses, while this site did not appear to be associated with maternal perceived stress during pregnancy in male-only samples. In sex-stratified SKAT models, prenatal perceived stress was most strongly associated with aggregate methylation in *FKBP5* in females and *HSD11B2* in males, though the association in males was only suggestive. While we found no prior evidence in the literature that showed that newborn *FKBP5* methylation differences in response to prenatal maternal stress is more pronounced in female compared to male

newborns, Appleton et al. [61] similarly showed that male infants had greater decreases in placental *HSD11B2* mean methylation in response to lower maternal education, increased poverty, and increased maternal socioeconomic risk than female infants. Conversely, Sutherland et al. [62] showed that in girls only with low levels of placental DNAm in *HSD11B2*, increased prenatal maternal depressive symptoms were associated with increased infant baseline cortisol. Sex differences in newborn DNAm changes associated with maternal prenatal stress exposure are largely understudied. Evidence has highlighted sex differences in foetal development and suggests an increased susceptibility to an adverse early life environment among male compared to female fetuses [63]. Future work is needed to better evaluate the presence and mechanisms of timing and sex differences in newborn DNAm patterns in response to the prenatal environment.

The relatively large sample size ($N = 661$) and the inclusion of participants from traditionally underrepresented racial/ethnic groups are among the main strengths of this study, leading to gains in power and improved generalizability of study results to the general US population. Additionally, our relatively novel use of SKAT in gene-level tests allowed for more flexible modelling and gains in power that better enabled us to detect the aggregate effect of DNAm in CpG within a region over burden tests that has been used in previous studies.

This study also has some important limitations. Primarily, while we did our best to control for confounders, it is possible that other unmeasured covariates, which have not been accounted for in our models, may play some role in the association between prenatal perceived stress and newborn DNAm in glucocorticoid related genes. Measurement error may have been another limitation of this study, particularly for self-reported measures such as maternal alcohol consumption or tobacco use during pregnancy, potentially leading to biased estimates. Additionally, it is possible that the association between maternal perceived stress during pregnancy and newborn DNAm may be modified by self-reported race, as experiences of stress among pregnant women have been shown to differ across sociodemographic categories, including race and ethnicity [35, 36]. However, given sample size limitations, we lacked the power to perform these sub-analyses. However, the primary goal of these analyses was to identify DNAm differences across racial/ethnic groups and so future well-powered studies that evaluate racial/ethnic differences may be valuable. It is also possible that the association varied across ECHO cohort sites, particularly given the difference in exposure distribution across sites. Due to insufficient sample sizes and inadequate power, we were unable to conduct cohort-specific analyses and elucidate potential cohort effects. However, as cohort site coincided with technical batch, batch correction may have minimized the effect of between cohort differences on our results.

This study and others have largely focused on the role of maternal experiences of acute SLEs and stress appraisal during pregnancy or perceived stress. We compared aggregate DNAm analysis methods, specifically mean DNAm burden tests and our SKAT-based approach. Our results suggest key advantages of the SKAT method, including gains in power that allowed us to identify statistically significant associations between prenatal perceived stress and newborn DNAm in *NR3C1* and *FKBP5*. These results suggest that future studies seeking to examine aggregate gene methylation associations with exposures or health outcomes would benefit from the use of SKAT over more traditional mean DNAm burden tests.

Future work should also explore other measures of stress, including cumulative or chronic stressors or biological stress correlates, such as cortisol, to better model the biological pathway between prenatal stress and offspring DNAm. Additionally, this study established an association between maternal perceived stress during pregnancy and newborn DNAm HPA axis-associated genes. As prior work has demonstrated connections between maternal perceived stress during pregnancy and offspring health outcomes, future work that evaluates whether the DNAm changes we identified mediate specific perinatal and child health outcomes is necessary. Finally, future studies should explore the potential impact of postnatal factors on offspring DNAm in order to identify potential intervention measures.

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Author contributions

Krystin Jones (Conceptualization [equal], Formal Analysis [equal], Visualization [lead], Writing – original draft [lead], Writing – review & editing [lead]), Bianca P. Acevedo (Writing – review & editing [supporting]), Lyndsay A. Avalos (Writing – review & editing [supporting]), Brennan H. Baker (Writing – review & editing [supporting]), Nicole R. Bush (Methodology [supporting], Writing – review & editing [supporting]), Claudia Buss (Writing – review & editing [equal]), Luke P. Grosvenor (Writing – review & editing [supporting]), Alison E. Hipwell (Writing – review & editing [supporting]), Kristine Marceau (Writing – review & editing [supporting]), Cindy T. McEvoy (Writing – review & editing [supporting]), Wei Perng (Writing – review & editing [supporting]), Alexandra D. W. Sullivan (Writing – review & editing [supporting]), Irene Tung (Writing – review & editing [supporting]), Yeyi Zhu (Writing – review & editing [supporting]), Christine Ladd-Acosta (Conceptualization [equal], Supervision [lead], Writing – original draft [supporting], Writing – review & editing [supporting])

Supplementary data

Supplementary data is available at [EnvEpig](#) online.

Conflict of interest : Dr. Ladd-Acosta reports consulting fees from the University of Iowa for providing expertise on autism epigenetics, outside the scope of this work. All other co-authors declare no conflicts of interest.

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Data availability

Select de-identified data from the ECHO Program are available through NICHD's Data and Specimen Hub (DASH). Information on study data not available on DASH, such as some Indigenous datasets, can be found on the ECHO study DASH webpage.

Appendix

Table A.1. Overview of Notation.

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Wei	Perng		PhD, MPH	Lifecourse Epidemiology of Adiposity and Diabetes (LEAD) Center	University of Colorado Anschutz Medical Campus	Aurora, Colorado, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023248 (Dana Dabelea)	wei.perng@cuanschutz.edu
Traci A.	Bekelman		PhD, MPH	Lifecourse Epidemiology of Adiposity and Diabetes (LEAD) Center	University of Colorado Anschutz Medical Campus	Aurora, Colorado, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023248 (Dana Dabelea)	traci.bekelman@cuanschutz.edu
Greta	Wilkening		PhD, MPH	Lifecourse Epidemiology of Adiposity and Diabetes (LEAD) Center	University of Colorado Anschutz Medical Campus	Aurora, Colorado, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023248 (Dana Dabelea)	Greta.Wilkening@childrenscolorado.org
Sheryl	Magzamen		PhD	Environmental and Radiological Health Sciences	Colorado School of Public Health, Colorado State University	Fort Collins, Colorado, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035526 (Dana Dabelea)	Sheryl.Magzamen@colostate.edu
Brianna F.	Moore		PhD, MS	Lifecourse Epidemiology of Adiposity and Diabetes (LEAD) Center	University of Colorado Anschutz Medical Campus	Aurora, Colorado, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035526 (Dana Dabelea)	brianna.f.moore@cuanschutz.edu
Anne P.	Starling		PhD	Epidemiology	University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035526 (Dana Dabelea)	anne.starling@unc.edu
Deborah J.	Rinehart		PhD	Center for Health Systems Research	Denver Health and Hospital Authority	Denver, Colorado, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035526 (Dana Dabelea)	deborah.rinehart@dhha.org
Daphne	Koinis Mitchell		Ph.D	Department of Pediatrics	Rhode Island Hospital, The Alpert Medical School of Brown University	Providence, Rhode Island, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023313 (Daphne Koinis Mitchell)	dkoinismitchell@lifespan.org
Viren	D'Sa		MD	Department of Pediatrics	Rhode Island Hospital, The Alpert Medical School of Brown University	Providence, Rhode Island, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023313 (Daphne Koinis Mitchell)	viren_Dsa@brown.edu
Sean C.L.	Deoni		PhD	Division of Gender Equality, Maternal, Newborn & Child Health Discovery & Tools Team	Bill & Melinda Gates Foundation	Seattle, Washington, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023313 (Daphne Koinis Mitchell)	Sean.Deoni@gatesfoundation.org

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Number and Grant Number	Email Address
Hans-Georg	Mueller		PhD	Department of Statistics	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site	UG3/UH3OD023313 (Daphne Koinis Mitchell)	hgmuel@ucdavis.edu
Cristiane S.	Duarte		PhD, MPH	Division of Child and Adolescent Psychiatry	Columbia University - NYSPi	New York, New York, USA	Co-Investigator ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	Cristiane.Duarte@nyspi.columbia.edu
Catherine	Monk		PhD	Department of Obstetrics & Gynecology	Columbia University - NYSPi	New York, New York, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	cem31@cumc.columbia.edu
Glorisa	Canino		PhD	Behavioral Sciences Research Institute	University of Puerto Rico, School of Medicine	Rio Piedras, Puerto Rico	ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	glorisa.canino@upr.edu
Jonathan	Posner		MD	Child & Family Mental Health & Community Psychiatry Division	Duke University School of Medicine, Duke Psychiatry & Behavioral Sciences	Durham, North Carolina, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	jonathan.posner@duke.edu
Tenneill	Murray		MPH	Division of Child and Adolescent Psychiatry	Columbia University - NYSPi	New York, New York, USA	ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	tenneill.murray@nyspi.columbia.edu
Claudia	Lugo-Candelas		PhD	Division of Child and Adolescent Psychiatry	Columbia University - NYSPi	New York, New York, USA	Co-Director ECHO Cohort Study Site Principal	UH3OD023328 (Cristiane Duarte)	claudia.lugo@nyspi.columbia.edu
Anne L.	Dunlop		MD, MPH	Department of Gynecology and Obstetrics	Emory University School of Medicine	Atlanta, Georgia, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023318 (Anne Dunlop)	amlang@emory.edu
Patricia A.	Brennan		PhD	Department of Psychology	Emory University	Atlanta, Georgia, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023318 (Anne Dunlop)	pbren01@emory.edu
Christine	Hockett		PhD	N/A; Department of Pediatrics	Avera Research Institute; University of South Dakota School of Medicine	Rapid City, South Dakota, USA; Sioux Falls, South Dakota, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023279 (Amy Elliott)	christine.hockett@avera.org
Amy	Elliott		PhD	N/A; Department of Pediatrics	Avera Research Institute ; University of South Dakota School of Medicine	Sioux Falls, South Dakota, USA	ECHO Cohort Study Site Principal	UG3/UH3OD023279 (Amy Elliott)	amyelliott@avera.org

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Assiamira	Ferrara		MD, PhD	Division of Research	Kaiser Permanente Northern California	Oakland, California, USA	ECHO Cohort Study Site Principal	UG3/UH3OD023289 (Assiamira Ferrara)	assiamira.ferrara@kp.org
Lisa A.	Croen		PhD	Division of Research	Kaiser Permanente Northern California	Oakland, California, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023342 (Kristen Lyall), UG3/UH3OD023289 (Assiamira Ferrara)	Lisa.A.Croen@kp.org
Monique M.	Hedderson		PhD	Division of Research	Kaiser Permanente Northern California	Oakland, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023289 (Assiamira Ferrara), UG3OD035540 (Monique Marie Hedderson)	Monique.M.Hedderson@kp.org
John	Ainsworth		PhD	Centre for Health Informatics	University of Manchester	Manchester, United Kingdom	ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern)	John.Ainsworth@manchester.ac.uk
Leonard B.	Bacharier		MD	Department of Pediatrics, Monroe Carell Jr Children's Hospital at Vanderbilt	Vanderbilt University Medical Center	Nashville, Tennessee, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern)	leonard.bacharier@vumc.org
Casper G.	Bendixsen		PhD	National Farm Medicine Center	Marshfield Clinic Research Institute	Marshfield, Wisconsin, USA	ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern)	Bendixsen.casper@marshfieldresearch.org
James E.	Gern		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern), UG3OD035509 (Anne Marie Singh)	gern@medicine.wisc.edu
Diane R.	Gold		MD	The Channing Division of Network Medicine; Department of Medicine	Brigham and Women's Hospital; Harvard Medical School	Boston, Massachusetts, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern)	redrg@channing.harvard.edu
Tina V.	Hartert		MD, MPH	Division of Pediatric Allergy, Immunology, and Pulmonary Medicine, Department of Medicine, Department of Pediatrics	Vanderbilt University Medical Center	Nashville, Tennessee, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern), UG3OD035516 and UG3OD035517 (Tina Hartert)	tina.hartert@vumc.org

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First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Daniel J.	Jackson		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	dj@medicine.wisc.edu
Christine C.	Johnson		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem), UG3OD035518 (Jennifer Straughen)	CJOHNSO1@hfhs.org
Christine L.M.	Joseph		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	gjoseph1@hfhs.org
Meyer	Kattan		MD	Department of Pediatrics	Columbia University Medical Center	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	mk2833@cumc.columbia.edu
Gurjit K.	Khurana Hershey		MD, PhD	Division of Asthma Research	Cincinnati Children's Hospital Medical Center	Cincinnati, Ohio, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	gurjit.hershey@cchmc.org
Robert F.	Lemanske, Jr.		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	lemanske@wisc.edu
Susan V.	Lynch		PhD	Department of Medicine	University of California	San Francisco, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	Susan.Lynch@ucsf.edu
Rachel L.	Miller		MD	Department of Medicine, Division of Clinical Immunology	Icahn School of Medicine at Mount Sinai	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem), UG3/UH3OD023290 (Julie Herbstman)	Rachel.miller2@mssm.edu
George T.	O'Connor		MD	Department of Pediatrics	Boston University School of Medicine	Boston, Massachusetts, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem)	goconnor@bu.edu
Carole	Ober		PhD	Department of Human Genetics	University of Chicago	Chicago, Illinois, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gem), UG3OD035509 (Anne Marie Singh)	c-ober@genetics.uchicago.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Dennis	Ownby		MD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal	UG3/UH3OD023282 (James Gern)	downby@augusta.edu
Katherine	Rivera-Spoljaric		MD	Department of Pediatrics	Washington University School of Medicine	St Louis, Missouri, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern), UG3OD035521 (Katherine Rivera-Spoljaric)	rivera_k@wustl.edu
Patrick H.	Ryan		PhD	Department of Pediatrics and College of Medicine; Division of Biostatistics and Epidemiology	University of Cincinnati	Cincinnati, Ohio, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern), UG3OD035509 (Anne Marie Singh)	patrick.ryan@cchmc.org
Christine M.	Seroogy		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern)	cmseroogy@wisc.edu
Anne Marie	Singh		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern), UG3OD035509 (Anne Marie Singh)	amsingh@wisc.edu
Robert A.	Wood		MD	Department of Pediatrics	Johns Hopkins University School of Medicine	Baltimore, Maryland, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern)	rwood@jhmi.edu
Edward M.	Zoratti		MD	Division of Allergy and Clinical Immunology	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023282 (James Gern), UG3OD035518 (Jennifer Straughen)	ezoratti@hfhs.org
Rima	Habre		ScD, MSc	Department of Population and Public Health Sciences	University of Southern California	Los Angeles, California, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023287 (Carrie Breton)	habre@usc.edu
Shohreh	Farzan		PhD	Department of Population and Public Health Sciences	University of Southern California	Los Angeles, California, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023287 (Carrie Breton)	sffarzan@usc.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Frank D.	Gilliland		MD, MPH, PhD	Department of Population and Public Health Sciences	University of Southern California	Los Angeles, California, USA	ECHO Cohort Study Site Principal Investigator	UH3OD0023287 (Carrie Breton)	gillilan@usc.edu
Irva	Hertz-Picciotto		PhD	MIND Institute and Department of Public Health Sciences	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023365 (Irva Hertz-Picciotto), UG3OD035550 (Rebecca Schmidt)	irher@ucdavis.edu
Deborah H.	Bennett		Ph.D	Department of Public Health Sciences	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023365 (Irva Hertz-Picciotto), UG3OD035550 (Rebecca Schmidt)	dhbennett@ucdavis.edu
Julie B.	Schweitzer		Ph.D	Department of Psychiatry and Behavioral Science and the MIND Institute	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023365 (Irva Hertz-Picciotto)	jschweitzer@ucdavis.edu
Rebecca J.	Schmidt		Ph.D	MIND Institute and Department of Public Health Sciences	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023365 (Irva Hertz-Picciotto), UG3/UH3OD023342 (Kristen Lyall), UG3OD035550 (Rebecca Schmidt)	rjschmidt@ucdavis.edu
Janine M.	LaSalle		PhD	Medical Microbiology and Immunology; MIND Institute	University of California, Davis	Davis, California, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023365 (Irva Hertz-Picciotto), UG3OD035550 (Rebecca Schmidt)	jmlasalle@ucdavis.edu
Alison E.	Hipwell		PhD, ClinPsyD	Psychiatry and Psychology	University of Pittsburgh	Pittsburgh, Pennsylvania, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023244 (Alison Hipwell)	hipwae@upmc.edu
Kate E.	Keenan		PhD	Psychiatry and Behavioral Neuroscience	University of Chicago	Chicago, Illinois, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023244 (Alison Hipwell)	keenan@uchicago.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Catherine J.	Karr		MD, MS, PhD	Department of Pediatrics, School of Medicine; Department of Environmental and Occupational Health Sciences; School of Public Health	University of Washington	Seattle, Washington, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 and UG3OD035528 (Catherine Karr)	ckarr@uw.edu
Nicole R.	Bush		PhD	Department of Psychiatry and Behavioral Sciences and Department of Pediatrics, School of Medicine	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr), UG3OD035519 (Qi Zhao)	nicole.bush@ucsf.edu
Kaja Z.	LeWinn		ScD	Department of Psychiatry and Behavioral Sciences, School of Medicine	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr), UG3OD035519 (Qi Zhao)	kaja.lewinn@ucsf.edu
Sheela	Sathyanarayana		MD, MPH	Department of Pediatrics, School of Medicine; Department of Environmental and Occupational Health Sciences, School of Public Health	University of Washington and Seattle Children's Research Institute	Seattle, Washington, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr), UG3OD035508 (Sheela Sathyanarayana)	sheela.sathyanarayana@seattlechildrens.org
Qi	Zhao		MD, PhD	Department of Preventive Medicine	University of Tennessee Health Science Center	Memphis, Tennessee, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr), UG3OD035519 (Qi Zhao)	qzhao11@uthsc.edu
Frances	Tylavsky		DrPH, MS	Department of Preventive Medicine	University of Tennessee Health Science Center	Memphis, Tennessee, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr)	ftyavsk@uthsc.edu
Kecia N.	Carroll		MD, MPH	Department of Pediatrics, Department of Environmental Medicine & Public Health	Icahn School of Medicine at Mount Sinai	New York, New York, USA	Investigator ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr), UG3/UH3OD023337 (Rosalind Wright)	kecia.carroll@mssm.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Christine T.	Loftus		MS MPH PhD	Department of Environmental and Occupational Health Sciences; School of Public Health	University of Washington	Seattle, Washington, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023271 (Catherine Karr)	cloftus@uw.edu
Leslie D.	Leve		PhD	Department of Counseling Psychology and Human Services & Prevention Science Institute	University of Oregon	Eugene, Oregon, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023389 (Leslie Leve)	leve@uoregon.edu
Jody M.	Ganiban		PhD	Department of Psychological and Behavioral Sciences	George Washington University	Washington, DC, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023389 (Leslie Leve)	ganiban@gwu.edu
Jenae M.	Neiderhiser		PhD	Department of Psychology	Penn State University	University Park, Pennsylvania, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023389 (Leslie Leve)	jenaemn@psu.edu
Scott T.	Weiss		MD	Channing Division of Network of Medicine, Department of Medicine	Brigham and Women's Hospital and Harvard Medical School	Boston, Massachusetts, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023268 (Scott Weiss)	scott.weiss@channing.harvard.edu
Augusto A.	Litonjua		MD	Pediatric Pulmonary Division, Department of Pediatrics	Golisano Children's Hospital, University of Rochester	Rochester, New York, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023268 (Scott Weiss)	augusto_litonjua@urmc.rochester.edu
Cindy T.	McEvoy		MD, MCR	Division of Neonatology, Department of Pediatrics	Oregon Health & Science University	Portland, Oregon, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023288 (Cynthia McEvoy)	mcevoyc@ohsu.edu
Eliot R.	Spindel		MD, PhD	Division of Neuroscience	Oregon National Primate Research Center	Beaverton, Oregon, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023288 (Cynthia McEvoy)	spindele@ohsu.edu
Robert S.	Tepper		MD, PhD	Division of Pediatric Pulmonology, Department of Pediatrics	Indiana School of Medicine	Indianapolis, Indiana, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023288 (Cynthia McEvoy)	rtepper@iu.edu

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Craig J.	Newschaffer		PhD	College of Health and Human Development	Penn State	State College, Pennsylvania, USA	ECHO Cohort Study Site Principal	UG3/UH3OD023342 (Kristen Lyall)	newschaffer@psu.edu
Kristen	Lyall		ScD	AJ Drexel Autism Institute	Drexel University	Philadelphia, Pennsylvania, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023342 (Kristen Lyall)	kld98@drexel.edu
Heather E.	Volk		PhD	Mental Health	Johns Hopkins University	Baltimore, Maryland, USA	Investigator ECHO Cohort Study Site Principal	UG3/UH3OD023342 (Kristen Lyall)	hvolk1@jhu.edu
Rebecca	Landa		PhD	Department of Psychiatry and Behavioral Sciences	Center for Autism and Related Disorders, Kennedy Krieger Institute, Johns Hopkins University	Baltimore, Maryland, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	landa@kennedykrieger.org
Sally	Ozonoff		PhD	MIND Institute, Department of Psychiatry	University of California Davis	Sacramento, California, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	sozonoff@ucdavis.edu
Joseph	Piven		MD	Department of Psychiatry	University of North Carolina	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	jpiven@med.unc.edu
Heather	Hazlett		PhD	Department of Psychiatry	University of North Carolina	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	heather_cody@med.unc.edu
Juhi	Pandey		PhD	Center for Autism Research	Children's Hospital of Philadelphia	Philadelphia, Pennsylvania, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	pandey@chop.edu
Robert	Schultz		PhD	Center for Autism Research	Children's Hospital of Philadelphia	Philadelphia, Pennsylvania, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	schultzrt@chop.edu
Steven	Dager		PhD	Department of Radiology	University of Washington	Seattle, Washington, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	srd@uw.edu
Kelly	Botteron		PhD	Department of Psychiatry	Washington University	St Louis, Missouri, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023342 (Kristen Lyall)	botteronk@wustl.edu

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First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Daniel	Messinger		PhD	Department of Psychology	University of Miami	Miami, Florida, USA	ECHO Cohort Study Site	UG3/UH3OD023342 (Kristen Lyall)	dmessinger@miami.edu
Wendy	Stone		PhD	Department of Psychology	University of Washington	Seattle, Washington, USA	Co-Investigator ECHO Cohort Study Site	UG3/UH3OD023342 (Kristen Lyall)	stonew@uw.edu
Jennifer	Ames		PhD	Kaiser Permanente Division of Research	Kaiser Permanente	Oakland, California, USA	Co-Investigator ECHO Cohort Study Site	UG3/UH3OD023342 (Kristen Lyall)	Jennifer.L.Ames@kp.org
Thomas G.	O'Connor		PhD	Departments of Psychiatry, Neuroscience, Obstetrics and Gynecology	University of Rochester	Rochester, New York, USA	Co-Investigator ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023349 (Thomas O'Connor)	tom_oconnor@urmc.Rochester.edu
Richard K.	Miller		PhD	Departments of Obstetrics and Gynecology	University of Rochester	Rochester, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023349 (Thomas O'Connor)	richardk_miller@urmc.rochester.edu
Emily	Oken		MD, MPH	Division of Chronic Disease Research Across the Lifecourse, Department of Population Medicine	Harvard Pilgrim Health Care Institute and Harvard Medical School	Boston, Massachusetts, USA	Investigator ECHO Cohort Study Site Principal Investigator	UH3OD023286 and UG3OD035533 (Emily Oken)	emily_oken@hms.harvard.edu
Michele R.	Hacker		ScD	Department of Obstetrics and Gynecology	Beth Israel Deaconess Medical Center	Boston, Massachusetts, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035533 (Emily Oken)	mhacker@bidmc.harvard.edu
Tamarra	James-Todd		PhD	Department of Environmental Health	Harvard Chan School of Public Health	Boston, Massachusetts, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035533 (Emily Oken)	tjtodd@hsph.harvard.edu
T. Michael	O'Shea	Jr	MD, MPH	Division of Neonatology, Department of Pediatrics	University of North Carolina School of Medicine	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023348 (Mike O'Shea), UH3OD023347 (Barry Lester)	moshea52@email.unc.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Rebecca C.	Fry		PhD	Department of Environmental Sciences and Engineering	University of North Carolina Gillings School of Global Public Health	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023348 (Mike O'Shea)	rfry@unc.edu
Jean A.	Frazier		MD	EK Shriver Center and Psychiatry	UMASS Chan Medical School	Worcester, Massachusetts, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	jean.frazier@umassmed.edu
Rachana	Singh		MD, MS	Department of Pediatrics	Tufts University School of Medicine	Boston, Massachusetts, USA	Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	jean.frazier@umassmed.edu
Caitlin	Rollins		MD, SM	Department of Neurology	Harvard Medical School	Boston, Massachusetts, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	Rachana.Singh1@tufsmmedicine.org
Angela	Montgomery		MD	Division of Neonatology, Department of Pediatrics	Yale School of Medicine	New Haven, Connecticut, USA	Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	angela.montgomery@yale.edu
Ruben	Vaidya		MD	Department of Pediatrics	University of Massachusetts Chan Medical School-Baystate	Springfield, Massachusetts, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	Ruben.VaidyaMD@baystatehealth.org
Robert M.	Joseph		PhD	Department of Anatomy & Neurobiology	Boston University Chobanian & Avedisian School of Medicine	Boston, Massachusetts, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	rmjoseph@bu.edu
Lisa K.	Washburn		MD	Pediatrics	Wake Forest School of Medicine	Winston-Salem, North Carolina, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	lwcadmus@gmail.com
Semsa	Gogcu		MD, MPH	Section of Neonatology, Department of Pediatrics; Department of Pediatrics	Wake Forest School of Medicine; Wake Forest University School of Medicine/Atrium Health Wake Forest	Winston-Salem, North Carolina, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea), UG3OD035513 (Annemarie Stroustrup), UH3OD023320 (Judy Aschner)	sgogcu@wakehealth.edu
Kelly	Bear		DO	Section of Neonatology, Department of Pediatrics	ECU Health	Greenville, North Carolina, USA	ECHO Cohort Study Site	UG3/UH3OD023348 (Mike O'Shea)	BEARK17@ECU.EDU

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Julie V.	Rollins		MA	Division of Neonatology, Department of Pediatrics	University of North Carolina School of Medicine	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Award Project Director	UG3/UH3OD023348 (Mike O'Shea)	julie.rollins@unc.edu
Stephen R.	Hooper		PhD	Department of Health Sciences	School of Medicine, University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	stephen_hooper@med.unc.edu
Genevieve	Taylor		MD	Pediatrics	School of Medicine, University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	gtaylor@med.unc.edu
Wesley	Jackson		MD, MPH	Division of Neonatology, Department of Pediatrics	University of North Carolina School of Medicine	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	wesley.jackson@unc.edu
Amanda	Thompson		PhD	Department of Anthropology, Department of Nutrition	University of North Carolina School of Medicine	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	althompson@email.unc.edu
Julie	Daniels		PhD	Epidemiology and Maternal and Child Health	University of North Carolina at Chapel Hill; Gillings School of Global Public Health, University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	julie_daniels@unc.edu
Michelle	Hernandez		MD	Pediatrics	School of Medicine, University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	michelle_hernandez@med.unc.edu
Kun	Lu		PhD	Environmental Sciences and Engineering	Gillings School of Global Public Health, University of North Carolina at Chapel Hill	Chapel Hill, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	kunlu@unc.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Michael	Msall		MD	Kennedy Research Center on Intellectual and Neurodevelopmental Disabilities	University of Chicago Medicine: Comer Children's Hospital	Chicago Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	mmsall@peds.bsd.uchicago.edu
Madeleine	Lenski		MSPH	Department of Epidemiology and Biostatistics	Michigan State University	East Lansing, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	lenskim@msu.edu
Rawad	Obeid		MD	Pediatrics	Beaumont Hospital	Royal Oak, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	Rawad.Obeid@beaumont.org
Steven L.	Pastyrnak		PhD	Pediatrics	Corewell Health, Helen DeVos Children's Hospital	Grand Rapids, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea), UH3OD023347 (Barry Lester)	Steve.Pastyrnak@helendevoschildrens.org
Elizabeth	Jensen		PhD	Epidemiology and Prevention	Wake Forest University School of Medicine	Winston-Salem, North Carolina, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	ejensen@wakehealth.edu
Christina	Sakai		MD	Pediatrics	Mass General Hospital for Children	Boston, Massachusetts, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023348 (Mike O'Shea)	Christina.sakai@gmail.com
Hudson	Santos		RN, PhD	Dean's Office Graduate School, School of Nursing and Health Studies	University of Miami	Coral Gables, Florida, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023348 (Mike O'Shea), UG3OD035542 (Hudson Santos)	hsantos@miami.edu
Jean M.	Kerver		PhD, MSc, RD	Departments of Epidemiology & Biostatistics, and Pediatrics & Human Development	Michigan State University, College of Human Medicine	East Lansing, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver)	kerverje@msu.edu
Nigel	Paneth		MD, MPH	Departments of Epidemiology & Biostatistics, and Pediatrics & Human Development	Michigan State University, College of Human Medicine	East Lansing, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver)	paneth@msu.edu
Charles J.	Barone	II	MD, FAAP	Department of Pediatrics	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver), UG3/UH3OD023282 (James Gern)	cbarone1@hfhs.org

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Michael R.	Elliott		PhD	Department of Biostatistics	University of Michigan	Ann Arbor, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver)	mrelliot@umich.edu
Douglas M.	Ruden		PhD	Department of Obstetrics and Gynecology; Institute of Environmental Health Sciences (IEHS), C.S. Mott Center for Human Health and Development	Wayne State University	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver)	douglasr@wayne.edu
Chris	Fussman		MS	Lifecourse Epidemiology and Genomics Division	Michigan Department of Health and Human Services (MDHHS)	Lansing, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023285 (Jean Kerver)	fussmanc@michigan.gov
Julie B.	Herbstman		PhD	Department of Environmental Health Sciences	Columbia University Mailman School of Public Health	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023290 (Julie Herbstman)	jh2678@cumc.columbia.edu
Amy	Margolis		PhD	Department of Psychiatry	Columbia University Irving Medical Center	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023290 (Julie Herbstman)	amy.margolis@nyspi.columbia.edu
Susan L.	Schantz		PhD	Beckman Institute for Advanced Science and Technology; Department of Comparative Biosciences	University of Illinois Urbana-Champaign	Urbana, Illinois, USA	Investigator ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023272 (Susan Schantz)	schantz@illinois.edu
Sarah Dee	Geiger		PhD	Beckman Institute for Advanced Science and Technology; Department of Kinesiology and Community Health	University of Illinois Urbana-Champaign	Urbana, Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023272 (Susan Schantz)	smurphy7@illinois.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, II)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Andrea	Aguiar		PhD	Beckman Institute for Advanced Science and Technology; Department of Comparative Biosciences	University of Illinois Urbana-Champaign	Urbana, Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023272 (Susan Schantz)	aaguiar@illinois.edu
Karen	Tabb		PhD, MSW	Beckman Institute for Advanced Science and Technology; Department of Social Work	University of Illinois Urbana-Champaign	Urbana, Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023272 (Susan Schantz)	ktabb@illinois.edu
Rita	Strakovsky		PhD	Department of Food Science and Human Nutrition	Michigan State University	East Lansing, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023272 (Susan Schantz)	strakovs@msu.edu
Tracey	Woodruff		PhD, MPH	Program on Reproductive Health and the Environment	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023272 (Susan Schantz)	tracey.woodruff@ucsf.edu
Rachel	Morello-Frosch		PhD, MPH	Department of Environmental Science, Policy and Management and School of Public Health	University of California, Berkeley	Berkeley, California, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023272 (Susan Schantz)	rmf@berkeley.edu
Amy	Padula		PhD	Program on Reproductive Health and the Environment	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023272 (Susan Schantz)	amy.padula@ucsf.edu
Joseph B.	Stanford		MD, MSPH	Department of Family and Preventive Medicine	Spencer Fox Eccles School of Medicine, University of Utah	Salt Lake City, Utah, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023249 (Joseph Stanford)	joseph.stanford@utah.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Christina A.	Porucznik		PhD, MSPH	Department of Family and Preventive Medicine	Spencer Fox Eccles School of Medicine, University of Utah	Salt Lake City, Utah, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023249 (Joseph Stanford)	christy.porucznik@utah.edu
Angelo P.	Giardino		MD, PhD	Department of Pediatrics	Spencer Fox Eccles School of Medicine, University of Utah	Salt Lake City, Utah, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023249 (Joseph Stanford)	giardino@hsc.utah.edu
Rosalind J.	Wright		MD, MPH	Department of Environmental Medicine & Public Health	Icahn School of Medicine at Mount Sinai	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023337 (Rosalind Wright)	rosalind.wright@mssm.edu
Robert O.	Wright		MD, MPH	Department of Environmental Medicine & Public Health	Icahn School of Medicine at Mount Sinai	New York, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023337 (Rosalind Wright)	robert.wright@mssm.edu
Brent	Collett		PhD	Department of Psychiatry and Behavioral Medicine	University of Washington, Seattle Children's Research Institute	Seattle, Washington, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035508 (Sheela Sathyanarayana)	brent.collett@seattlechildrens.org
Nicole	Baumann-Blackmore		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	nlbaumann@wisc.edu
Ronald	Gangnon		PhD	Department of Population Health Sciences	University of Wisconsin	Madison, Wisconsin, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	ronald@biostat.wisc.edu
Daniel J.	Jackson		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	djj@medicine.wisc.edu
Chris G.	McKennan		PhD	Department of Statistics	University of Pittsburgh	Pittsburgh, Pennsylvania, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	CHM195@pitt.edu
Jo	Wilson		MD	Department of Pediatrics	University of Wisconsin School of Medicine and Public Health	Madison, Wisconsin, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	wilson54@wisc.edu
Matt	Altman		MD	Department of Medicine	University of Washington	Seattle, Washington, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035509 (Anne Marie Singh)	maltman@benaroyresearch.org

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Judy L.	Aschner		MD	Department of Pediatrics	Albert Einstein College of Medicine; Hackensack Meridian School of Medicine; Center for Discovery and Innovation	Bronx, New York, USA; Nutley, New Jersey, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023320 and UG3OD035546 (Judy Aschner), UG3OD035513 (Annemarie Stroustrup)	judyaschner@einsteinmed.edu ; judyaschner@hmn.org
Annemarie	Stroustrup		MD, MPH	Department of Pediatrics	Northwell Health, Cohen Children's Medical Center, and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Principal Investigator	UH3OD023320 (Judy Aschner), UG3OD035513 (Annemarie Stroustrup)	astroustrup@northwell.edu
Stephanie L.	Merhar		MD, MS	Department of Pediatrics	Cincinnati Children's	Cincinnati, Ohio, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner), UG3OD035513 (Annemarie Stroustrup)	stephanie.merhar@cchmc.org
Paul E.	Moore		MD	Department of Pediatrics	Vanderbilt University Medical Center	Nashville, Tennessee, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner), UG3OD035513 (Annemarie Stroustrup)	paul.moore@vumc.org
Gloria S.	Pryhuber		MD	Department of Pediatrics	University of Rochester Medical Center	Rochester, New York, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner)	gloria_pryhuber@urmc.rochester.edu
Mark	Hudak		MD	Department of Pediatrics	University of Florida College of Medicine	Jacksonville, Florida, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner)	mark.hudak@jax.ufl.edu
Ann Marie	Reynolds Lyndaker		MD, MPH	Department of Pediatrics	University of Buffalo Jacobs School of Medicine and Biomedical Sciences	Buffalo, New York, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner)	amr1@buffalo.edu
Andrea L.	Lampland		MD	Department of Pediatrics	Children's Minnesota	Minneapolis, Minnesota, USA	ECHO Cohort Study Site Co-Investigator	UH3OD023320 (Judy Aschner)	andrea.lampland@childrensmns.org
Burton	Rochelson		MD	Department of Obstetrics and Gynecology	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035532 (Annemarie Stroustrup)	brochels@northwell.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Sophia	Jan		MD, MSHP	Department of Pediatrics	Northwell Health, Cohen Children's Medical Center, and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	sjan1@northwell.edu
Matthew J.	Blitz		MD, MBA	Department of Obstetrics and Gynecology	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	mblitz@northwell.edu
Michelle W.	Katzow		MD, MS	Department of Pediatrics	Northwell Health, Cohen Children's Medical Center, and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	mkatzow@northwell.edu
Zenobia	Brown		MD, MPH	Department of Science Education	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	zbrown2@northwell.edu
Codruta	Chiuzan		PhD	Institute of Health System Science	Northwell Health, Feinstein Institutes for Medical Research	Manhasset, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	cchiuzan@northwell.edu
Timothy	Rafael		MD	Department of Obstetrics and Gynecology	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	trafael@northwell.edu
Dawnette	Lewis		MD, MPH	Department of Obstetrics and Gynecology	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	dlewis@northwell.edu
Natalie	Meirowitz		MD	Department of Obstetrics and Gynecology	Northwell Health and the Zucker School of Medicine at Hofstra / Northwell	New Hyde Park, New York, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035532 (Annemarie Stroustrup)	nmeirowi@northwell.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Brenda	Poindexter		MD	Department of Pediatrics	Children's Healthcare of Atlanta Emory University	Atlanta, Georgia, USA	ECHO Cohort Study Site	UH3OD023320 (Judy Aschner)	breda.poindexter@emory.edu
Tebeb	Gebretsadik		MPH	Department of Biostatistics	Vanderbilt University Medical Center	Nashville, Tennessee, USA	Co-Investigator ECHO Cohort Study Site Principal Investigator	UG3OD035516 and UG3OD035517 (Tina Hartert)	tebeb.gebretsadik@vumc.org
Sarah	Osmundson		MD, MSC	Department of Obstetrics and Gynecology	Vanderbilt University Medical Center	Nashville, Tennessee, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035517 (Tina Hartert)	sarah.osmundson@vumc.org
Jennifer K.	Straughen		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035518 (Jennifer Straughen)	jstraug1@hfhs.org
Amy	Eapen		MD	Division of Allergy and Clinical Immunology	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035518 (Jennifer Straughen)	aeapen1@hfhs.org
Andrea	Cassidy-Bushrow		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023282 (James Gern)	acassid1@hfhs.org
Ganesa	Wegienka		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023282 (James Gern)	gwegien1@hfhs.org
Alex	Sitarik		MPH	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Biostatistician	UG3/UH3OD023282 (James Gern)	asitar1@hfhs.org
Kim	Woodcroft		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Co-Investigator	UG3OD035518 (Jennifer Straughen), UG3/UH3OD023282 (James Gern)	kwoodcr1@hfhs.org
Audrey	Urquhart		MPH	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site Epidemiologist	UG3OD035518 (Jennifer Straughen), UG3/UH3OD023282 (James Gern)	aurquha1@hfhs.org

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Albert	Levin		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	ECHO Cohort Study Site	UG3OD035518 (Jennifer Straughen)	alevin1@hfhs.org
Tisa	Johnson-Hooper		MD	Department of Pediatrics	Henry Ford Health	Detroit, Michigan, USA	Co-Investigator ECHO Cohort Study Site	UG3OD035518 (Jennifer Straughen)	tjohnso2@hfhs.org
Brent	Davidson		MD	Department of Women's Health	Henry Ford Health	Detroit, Michigan, USA	Co-Investigator ECHO Cohort Study Site	UG3/UH3OD023282 (James Gern)	bdavids1@hfhs.org
Tengfei	Ma		PhD	Department of Public Health Sciences	Henry Ford Health	Detroit, Michigan, USA	Co-Investigator ECHO Cohort Study Site	UG3OD035518 (Jennifer Straughen)	tengfei.ma@hfhs.org
Emily S.	Barrett		PhD	Department of Biostatistics and Epidemiology	Environmental and Occupational Health Sciences Institute, Rutgers University	Piscataway, New Jersey, USA	Co-Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	emily.barrett@eoshi.rutgers.edu
Martin J.	Blaser		MD	Center for Advanced Biotechnology & Medicine	Rutgers University	Piscataway, New Jersey, USA	Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	blaser@cabm.rutgers.edu
Maria Gloria	Dominguez-Bello		PhD	Departments of Biochemistry and Microbiology & Anthropology	Rutgers University	New Brunswick, New Jersey, USA	Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	mg.dominguez-bello@rutgers.edu
Daniel B.	Horton		MD	Department of Pediatrics	Robert Wood Johnson Medical School, Rutgers University	New Brunswick, New Jersey, USA	Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	daniel.horton@rutgers.edu
Manuel	Jimenez		MD	Departments of Pediatrics, Family Medicine, and Community Health	Robert Wood Johnson Medical School, Rutgers University	New Brunswick, New Jersey, USA	Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	jimenema@rwjms.rutgers.edu
Todd	Rosen		MD	Department of Obstetrics, Gynecology, and Reproductive Sciences	Robert Wood Johnson Medical School, Rutgers University	New Brunswick, New Jersey, USA	Co-Investigator ECHO Cohort Study Site	UG3OD035527 (Emily S Barrett)	rosentj@rwjms.rutgers.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Kristy	Palomares		MD, PhD	Department of Obstetrics and Gynecology	Saint Peter's University Hospital	New Brunswick, New Jersey, USA	ECHO Cohort Study Site	UG3OD0035527 (Emily S Barrett)	kpalomares@saintpetersuh.com
Lyndsay A.	Avalos		PhD, MPH	Division of Research	Kaiser Permanente Northern California	Oakland, California, USA	Co-Investigator ECHO Cohort Study Site Principal Investigator	UG3OD0035540 (Monique Marie Hedderson)	Lyndsay.A.Avalos@kp.org
Yeyi	Zhu		PhD, MS	Division of Research	Kaiser Permanente Northern California	Oakland, California, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035540 (Monique Marie Hedderson)	Yeyi.Zhu@kp.org
Kelly J.	Hunt		PhD	Department of Public Health Sciences	Medical University of South Carolina	Charleston, South Carolina, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035543 (Kelly J Hunt)	huntke@muscc.edu
Roger B.	Newman		MD	Department of Obstetrics and Gynecology	Medical University of South Carolina	Charleston, South Carolina, USA	Investigator ECHO Cohort Study Site Principal Investigator	UG3OD0035543 (Kelly J Hunt)	newmanr@muscc.edu
Michael S.	Bloom		PhD	Department of Global and Community Health	George Mason University	Fairfax, Virginia, USA	Investigator ECHO Cohort Study Site Principal Investigator	UG3OD0035543 (Kelly J Hunt)	mbloom22@gmu.edu
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Heather H.	Burris		MD, MPH	Division of Neonatology, Department of Pediatrics	Children's Hospital of Philadelphia; University of Pennsylvania Perelman School of Medicine	Philadelphia, Pennsylvania, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035537 (Sunni L Mumford)	BURRISH@chop.edu

Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Sara B.	DeMauro		MD, MSCE	Division of Neonatology, Department of Pediatrics	Children's Hospital of Philadelphia; University of Pennsylvania Perelman School of Medicine	Philadelphia, Pennsylvania, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035537 (Sunni L Mumford)	DEMAURO@chop.edu
Lynn M.	Yee		MD, MPH	Division of Maternal-Fetal Medicine, Department of Obstetrics & Gynecology	Feinberg School of Medicine, Northwestern University	Chicago, Illinois, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035546 (Judy Aschner)	lynn.yee@northwestern.edu
Aaron	Hamvas		MD	Division of Neonatology, Department of Pediatrics	Ann & Robert H. Lurie Children's Hospital, Feinberg School of Medicine, Northwestern University	Chicago, Illinois, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035546 (Judy Aschner)	ahamvas@luriechildrens.org
Antonia F.	Olidipo		MD, MSCI	Division of Maternal-Fetal Medicine, Department of Obstetrics & Gynecology	Hackensack University Medical Center, Hackensack Meridian School of Medicine	Nutley, New Jersey, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035546 (Judy Aschner)	antonia.olidipo@hmn.org
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Kirin N.	Suri		MD	Division of Developmental and Behavioral Pediatrics, Department of Pediatrics	Hackensack University Medical Center, Hackensack Meridian School of Medicine	Nutley, New Jersey, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035546 (Judy Aschner)	kirin.suri@hmn.org
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Raye-Ann O.	DeRegnier		MD	Division of Neonatology, Department of Pediatrics	Ann & Robert H. Lurie Children's Hospital, Feinberg School of Medicine, Northwestern University	Chicago, Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035546 (Judy Aschner)	r-deregnier@northwestern.edu
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Meghan M.	JaKa		PhD, MS	Division of Research & Evaluation	HealthPartners Institute	Minneapolis, Minnesota, USA	ECHO site Principal Investigator	UG3OD0035529 (Hong-Ngoc Nguyen)	meghan.m.jaka@healthpartners.com
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Isabel	Maldonado		MPH, BS	School of Nursing and Health Studies	University of Miami Miller School of Medicine	Miami, Florida, USA	ECHO Cohort Study Site Program Director	UG3OD0035542 (Hudson Santos)	icm16@miami.edu
Meghan	Miller		PhD	Psychiatry and Behavioral Sciences; MIND Institute	University of California Davis	Sacramento, California, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035550 (Rebecca Schmidt)	mrhmiller@ucdavis.edu

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First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Jonathan L.	Slaughter		MD, MPH	Center for Perinatal Research, Abigail Wexner Research Institute and Division of Neonatology, Nationwide Children's Hospital and Department of Pediatrics, College of Medicine and Division of Epidemiology, College of Public Health, The Ohio State University	Nationwide Children's Hospital and The Ohio State University	Columbus, Ohio, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035536 (Jonathan Slaughter)	jonathan.slaughter@nationwidechildrens.org
Sarah A.	Keim		PhD, MS, MA	Center for Biobehavioral Health, Abigail Wexner Research Institute, Nationwide Children's Hospital and Department of Pediatrics, College of Medicine and Division of Epidemiology, College of Public Health, The Ohio State University	Nationwide Children's Hospital and The Ohio State University	Columbus, Ohio, USA	ECHO Cohort Study Site Principal Investigator	UG3OD035536 (Jonathan Slaughter)	Sarah.Keim@nationwidechildrens.org

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First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Courtney D.	Lynch		PhD, MPH	Division of Maternal-Fetal Medicine, Department of Obstetrics and Gynecology, College of Medicine and Division of Epidemiology, College of Public Health, The Ohio State University	The Ohio State University	Columbus, Ohio, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035536 (Jonathan Slaughter)	Courtney.Lynch@osumc.edu
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First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Thomas F.	Northrup		PhD	Department of Family and Community Medicine	University of Texas Health Science Center at Houston (UTHealth Houston) McGovern Medical School	Houston, Texas, USA	ECHO Cohort Study Site Principal Investigator	UG3OD0035544 (Kristina Whitworth)	thomas.f.northrup@uth.tmc.edu
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Ricardo A.	Mosquera		MD	Department of Pediatrics	University of Texas Health Science Center at Houston (UTHealth Houston) McGovern Medical School	Houston, Texas, USA	ECHO Cohort Study Site Co-Investigator	UG3OD0035544 (Kristina Whitworth)	ricardo.a.mosquera@uth.tmc.edu
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Juliette C.	Madan		MD, MS	Departments of Psychiatry, Pediatrics & Epidemiology	Geisel School of Medicine at Dartmouth, Dartmouth Hitchcock Medical Center	Hanover, New Hampshire, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023275 (Margaret Karagas)	juliette.madan@dartmouth.edu
Debra M.	Mackenzie		PhD	Community Environmental Health Program, Department of Pharmaceutical Sciences	College of Pharmacy, University of New Mexico Health Sciences Center	Albuquerque, New Mexico, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023344 (Debra Mackenzie)	dmackenzie@salud.unm.edu
Johnmye L.	Lewis		PhD	Community Environmental Health Program, Department of Pharmaceutical Sciences	College of Pharmacy, University of New Mexico Health Sciences Center	Albuquerque, New Mexico, USA	ECHO Cohort Study Site Principal Investigator	UG3/UH3OD023344 (Debra Mackenzie)	jlewis@cybermesa.com ; jlewis@salud.unm.edu
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, III)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Bennett L.	Leventhal		MD	Community Environmental Health Program, Department of Pharmaceutical Sciences UNM	College of Pharmacy, University of New Mexico Health Sciences Center, University of Chicago	Albuquerque, New Mexico, USA; Chicago, Illinois, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023344 (Debra MacKenzie)	Bennett.leventhal@outlook.com
Young Shin	Kim		MD, MS, MPH, PhD	Department of Psychiatry and Behavioral Sciences	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023344 (Debra MacKenzie)	Youngshin.Kim@ucsf.edu
Somer	Bishop		PhD	Department of Psychiatry and Behavioral Sciences	University of California, San Francisco	San Francisco, California, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023344 (Debra MacKenzie)	Somer.Bishop@ucsf.com
Sara S.	Nozadi		PhD	Community Environmental Health Program, Department of Pharmaceutical Sciences	College of Pharmacy, University of New Mexico Health Sciences Center	Albuquerque, New Mexico, USA	ECHO Cohort Study Site Co-Investigator	UG3/UH3OD023344 (Debra MacKenzie)	snozadi@unm.edu
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Table A.1. Continued

First Name and Middle Initial(s)	Last Name	Suffix (e.g., Jr, II)	Academic Degrees	Department	Institution	Location (city, state/province, country)	Role or Contribution	ECHO Cohort Study Site or Core Name and Grant Number	Email Address
Todd	Everson		PhD	Department of Environmental Health	Rollins School of Public Health, Emory University	Atlanta, Georgia, USA	ECHO Cohort Study Site Principal	UH3OD023347 (Barry Lester)	todd.m.everson@emory.edu
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Brian S.	Carter		MD	Department of Pediatrics	Children's Mercy-Kansas City	Kansas City, Missouri, USA	ECHO Cohort Study Site Principal	UH3OD023347 (Barry Lester)	bscarter@cmh.edu
Jennifer	Check		MD	Department of Pediatrics	Wake Forest School of Medicine	Winston, Salem North Carolina, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023347 (Barry Lester)	jcheck@wakehealth.edu
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Charles R.	Neal		MD	Department of Pediatrics	University of Hawaii John A Burns School of Medicine	Honolulu, Hawaii, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023347 (Barry Lester)	cneal@hphmg.org
Lynne M.	Smith		MD	Department of Pediatrics	UCLA Clinical and Translational Science Institute at The Lundquist Institute, Harbor-UCLA Medical Center	Los Angeles, California, USA	Investigator ECHO Cohort Study Site Principal	UH3OD023347 (Barry Lester)	smith@lundquist.org-test

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